

OEE Dashboard using Data Analytics

1st Ratnmala Bhimanpallewar
Department of Information Technology
BRAC's Vishwakarma Institute of
Information Technology
Pune, India
ratnmalab@gmail.com
ORCID: 0000-0001-8098-977

2nd Dhiraj Jagtap
Department of Information Technology
BRAC's Vishwakarma Institute of
Information Technology
Pune, India
dhiraj.22220027@viit.ac.in

3rd Vasudev Raut
Department of Information Technology
BRAC's Vishwakarma Institute of
Information Technology
Pune, India
vasudev.22220299@viit.ac.in

4th Pradeep Tonge
Crompton Experience and Innovation Center,
CGCEL,
Mumbai, India
pradeep.patil123@gmail.com

Abstract—To assess the productivity of the equipment, the overall equipment effectiveness, a crucial performance metric, is used (OEE). The goal of this work includes examining and analyzing OEE growth and identifying potential regions for future development. The usefulness of the current dashboards is discussed in this study along with a thorough analysis of the literature and cutting-edge analytical techniques. When analyzing equipment's utilization capacity and unit production, the individual stoppages and break times must be taken into consideration. The findings of OEE must also be changed accordingly. Correlation between technical and management difficulties is the fundamental strategy used to get the desired results. Analysis results are made available to the employee who look after the production statistics, here the results prepared using modern tool. The major difference in traditional and modern tool result visualization is, tools give more interactive analysis which can be easily interpreted by a non-technical person as well. Real-world efficiency has increased, but as technology develops, new opportunities for improvement frequently appear.

Keywords— OEE-overall equipment effectiveness component; Performance; Availability; Quality; CNC; KPI; Data Analytics

I. INTRODUCTION

Data computations, equipment performance evaluations, and estimating a company's output rate were all challenging tasks in the past. The business analytics functionality of the OEE (Overall Equipment Effectiveness) dashboard is an advancement over preceding technologies [1]. Continuously used production equipment will experience a decrease, necessitating maintenance. Regardless of the industry they operate in, modern businesses face challenges that are primarily shared by all of them and are largely centered on the explosion of data. An analytics dashboard is a highly visual representation of data that is used to track conditions or quickly aid comprehension. The origin of OEE measurement tool we can say the TPM (Total Productive Maintenance) concept which was brought by Nakajima (1988) [2]. Such dashboards are required to lower the likelihood of human error and delay while ensuring that the appropriate machinery is being used to meet production goals. Early alerts from OEE enable us to prevent downtime and make scheduled deliveries of items. The dashboard's primary function is to show the outcomes of calculations for cycle time, production time,

amount needed, and quantity achieved. This will reduce the time needed for unit production, enhance machine optimization, and help the industry assess the manufacturing of manufactured items. Using this dashboard, which will show a graphed overview of machine performance, the user will be able to forecast costs, profit, loss, production units etc.

II. RELATED WORK

To understand clearly how dashboards or statistical analysis helps in improving the manufacturing units we have discussed related systems briefly. One of the study outlines the procedure about BI solution development of one standard CNC machining centre. It was intended to combine data from CNC controllers of industrial machines through OPCUA in a potent unified repository in order to analyse all data and produce information on the status of the computer's activity [1]. The Grafana visualization tool for streaming static and time-series data was the main feature considered. A simulation tool is also available [3]. The methodology presented in this study consists of comprehending the real world, turning it into a conceptual model, using simulation to create a computer model, and then analysing that model to derive the foundation for OEE measurements using data analytics tools. The investment in software necessary to properly harness the power of the tools is one constraint of this method. The article explains how to configure Grafana with Influx DB and visualize data [4]. Along with pulling the data from the Influx DB, it also demonstrates how to construct a new Grafana Dashboard and a new Influx DB data source.

Here researcher has discussed about system that analyses data on urban traffic and effect of traffic emissions on urban air quality using information visualisation techniques[5]. Good visualisations make it possible for citizens and government agencies to spot patterns, identify crowded road portions at particular times, and maintain traffic sensor monitoring systems.

Data computations, equipment performance evaluations, and estimating a company's output rate were all challenging tasks in the past. The business analytics functionality of the OEE (Overall Equipment Effectiveness) dashboard is an advancement over preceding technologies. The Management of Modern equipment was different, it starts with preventive maintenance and updated with productive maintenance [3].

Many businesses place more emphasis on major time losses caused by breakdowns than on smaller time and speed losses.[6]. Managers or other senior employees of a company can use it with the push of a button to compute the company's profit and evaluate the availability of the equipment inside the organization. The term "business intelligence" refers to an expansive notion that includes the gathering, integrating, analyzing, and visualizing of corporate data with the intention of facilitating and improving decision-making.

III. OVERALL EQUIPMENT EFFECTIVENESS

The benchmark for assessing manufacturing productivity is OEE (Overall Equipment Effectiveness). In other words, it shows how much of the time spent manufacturing is actually productive. A 100% OEE score indicates that you are producing only Good Parts as quickly as possible without any Stop Time. That means 100% Quality (only Good Parts), 100% Performance (as quickly as possible), and 100% Availability in the OEE lingo (no Stop Time). A manufacturing best practice is measuring OEE. You will learn crucial lessons about how to methodically enhance your manufacturing process by measuring OEE and the underlying losses. The best indicator of losses, benchmarking development, and raising manufacturing equipment productivity is OEE (i.e., eliminating waste).

The fundamental goal of OEE is to monitor increases in industrial productivity (i.e., eliminating waste). Total Effective Equipment Performance (TEEP) is the ratio of Fully Productive Time to All Time (24/7). OEE dashboards are required to lower the likelihood of human mistake and delay and to make sure that the appropriate machinery is being used to accomplish production goals. Platforms for machine data, such as Machine Metrics, use data straight from the machine's control to more precisely measure OEE. As a result, all equipment manufacturers and models will have precise, consistent, and real-time production data [3].

Here we are going to discuss about theoretical foundation and investigation into the advantages of overall equipment effectiveness. Performance measurement is the framework in which OEE literature is created and reported. The analysis of (OEE) measurements comes next.

IV. METHODOLOGY

After the text edit has been completed, the paper is ready

To make this OEE dashboard, we have used the analytical tool i.e. Power BI powered by Microsoft. This analytical part was earlier implemented manually using Excel sheet. The most effective IoT implementation partners are digital technologies, which may increase automation, decrease mistakes, promote proactiveness, simplify company operations, improve knowledge-intensive procedures, lower costs, boost intelligence, and generally help people do more with less resources.

This can be achieved by using advanced analytical tools like Power BI. Power BI is the open-source analytics & monitoring solution for every database. For industry or other users, it can be visualized to get better understanding with the help of advanced queries and transformations.

The objective of this study is to create a BI system that makes the data uniform from various data sources to enable their treatment and internal and external knowledge production,

key performance indicator (KPIs) which further help for decision making, the emphasis here is majorly on:

- A. *Importing data files and classifying data are part of data preparation.*
- B. *Data modeling is the process of connecting data sets and generating derived metrics.*
- C. *invention of data visualization (i.e. data reports)*
- D. *Validation of data reports*

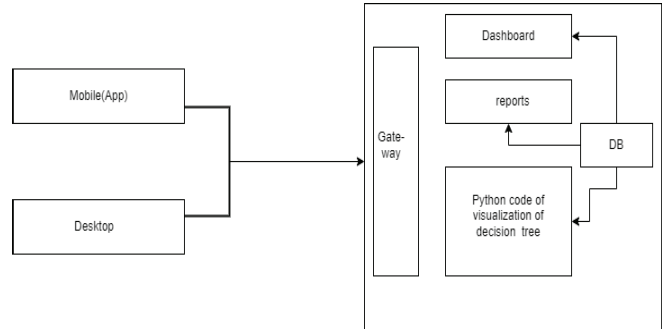


Fig. 1. Proposed System Architecture Working of the System

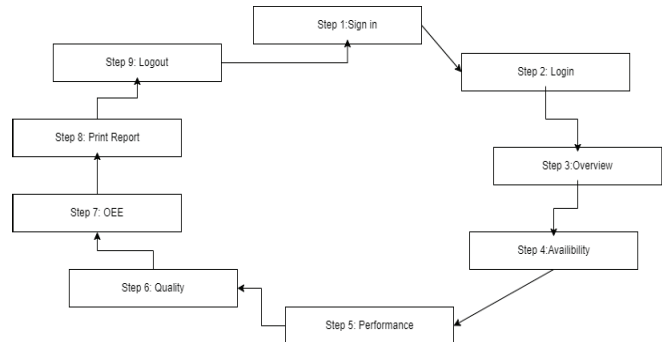


Fig. 2. Process Flow

Fig 1 shows the system architecture and Fig 2 represents process flow as discussed below step-wise

- **Login:** After signing in, the user will be redirected to the power bi login page.
- **Overview:** After signing in the user will be redirected to power bi dashboard. Overview of the data will be displayed.
- **Availability:** It shows the Availability calculation of the system with various visualizations and filters.
- **Performance:** It shows the Performance calculation of the system with various visualizations and filters.
- **Quality:** It shows the Quality calculation of the system with various visualizations and filters.
- **OEE:** It shows the OEE calculation of the system which helps the user take future production decisions.
- **Print Report:** Print report options enables the user to print the report of the production made in a day or month.
- **Logout:** If required, the user logout of the system.

A. OEE Monitoring Module

Data Acquisition:

The emphasis of this section is on: Given that the goal of this study is to develop an intelligent system get the data from every source uniformly so that the data is good enough to process and internal and external knowledge production (KPIs) to support the decision-making process [4].

Organizing and categorizing data for the OEE Monitoring Module:

Depending on the results of the subsequent tests, the various data collected from the data collection system are categorized. Calculating various OEE components experimentally, Overall effectiveness of equipment can be calculated as:

OEE = rate of quality, performability and availability.

1) *Availability:*

Operating time proportional to planned production Time is the same as availability. It indicates the period of the allotted time during which the machine may be available for use. Losses from downtime are considered.

Availability = Available time - unplanned downtime

Availability Time = Total Available Time - Planned downtime.

Planned downtime instances include many factors like team meetings, excess capacity, scheduled breaks, communication breaks and scheduled maintenance. Unplanned downtime can be brought on by breakdowns, configuration and adjustment mistakes, operator availability and delayed material deliveries.

2) *Performance:*

We can access the performance with the help of equation. A fundamental management principle is performance measurement. Performance measurement is crucial for managing the maintenance function, just like it is for other manufacturing functions [4].

Performance = (Total: Production-Parts/Running-Time) / Idle Run Rate.

Available time minus unplanned downtime equals operating time. Idle run rate is nothing but the number of parts produced per minute.

3) *Quality Rate:*

Manufacturers are increasingly required to provide information about the quality of their products[7]. The following equation is used to calculate the quality rate:

Quality rate= Total: produced-parts - defective parts

Statistics based methods are used in manufacturing domain. One of the methodologies is proposed based on profiles to methodically guide the manufacturing sector's choice and setting up ML-based analytics solutions[8]. It is also used in medical sectors [9].

B. *Data Processing and Modeling*

The data has been categorised and processed up to that point using "Query Editor" in accordance with the guidelines for creating a data modelling architecture. After the data had been processed, a relationship between each set of data and the others had been established, enabling an analysis of how one set of data affected the others.

The advanced visuals, which are made up of a variety of features and numerous analytical capabilities, will be used in the reports that list the current state of OEE. That stated as a Key Performance Indicator using the recently created dashboard.

C. *Dashboard with feature Visual Analytics*

Visualisations are "Published", in the form of service, is created dashboard includes the fields, for example data and transformed Visual analytics. In general, organisation members who are a part of the workspace may share the dashboard with anyone, but they cannot tamper with the data because there won't be a choice given to them. The visualisations can be viewed and reports can be generated by someone who is only allowed to use the dashboard. Access to the individuals who can edit the data in the dashboard after publishing can be granted by the user who created the workspace, but this is only possible for confidential users.

D. *The Components*

Device Monitoring modules available with app are hosted on the service webserver. Through web server one can create and share such reports. It is crucial to set the right values for machine and process-parameters at the start of the process of production to achieve highly efficient final products, here effectiveness in the terms of repeatability as well as quality [8,9]. According to the specifications to categorise, assess, and highlight the KPI of each module, dashboards have been created. The dashboard primarily consists of "three web pages/tabs" constructed with "Tiles" sections that display graphs and statistics. The three web pages/tabs that make up the platform are known as the module named Availability-Performance, further divided as module for

- 1) Quality Monitoring.
- 2) Overall equipment effectiveness
- 3) Monitoring Performance and Availability.

E. *Use of AI in Power BI:*

AI (Artificial Intelligence) -enabled Businesses can get important insights from unused data thanks to business intelligence (BI). Business intelligence tools with AI capabilities may examine fresh data and identify any important trends for the firm. In order to expand the scope of the information provided, AI also enables BI to benefit from cutting-edge technologies like predictive analytics, machine learning, and natural language processing. Instead of depending on a visual dashboard of big data trends, organizations seek solutions that bridge the gap between visual representation and actionable insights. Business intelligence enabled by AI can be useful in this situation.

We have made an effort to evaluate or report using Random Forest, a machine learning technique. In classification and regression issues, random forest and other supervised machine learning algorithms are frequently used. It creates decision trees from various samples, categorising them based on their average, and regressing them based on a majority vote. The best part of Random Forest Algorithm is, it has capacity to deal with variety of dataset having continuous attributes and categorical attributes as well. Regarding categorization issues, it yields better results [10].

In this dashboard we have used the Random Forest algorithm for analyzing our output for finding the major issues for the loss of the production. We have achieved the decision

making using the Weka tool which helped us to work on the dataset and visualize the tree representing the issue that causes most loss in the top i.e. setting time in our case and other issue in the order below. Weka takes the data, represents it in the form of visuals and represents the output or decision.

F. Machine Learning Algorithms used:

Machine Learning (ML) algorithms are broadly categorised into supervised and unsupervised algorithms. It works in two phases training and testing. For training and testing the dataset used is different and accuracy can be calculated through testing dataset. According to empirical data, the OOB (out of bounds) error for bagged classifiers is as accurate as using a test set with the same size as the training set [11].

The OOB estimate of the generalisation error is the classification error rate for all test sets. Consequently, using the OOB estimate does away with the need for a separate test set. Each individual CART (Classification And Regression Tree) tree casts a vote for one class, and the forest predicts the class that receives the majority of votes to categorise new input data [12]. Since many decades such predictions techniques are based on Machine learning (ML) [13] and used in medical domain as well [14].

G. Decision Tree:

Decision Trees comes under supervised classification approach [15]. The primary technique for categorization and prediction is decision trees. Decision tree learning is a common inductive approach based on examples, focusing on classification rules that show up as decision trees inferred from a collection of disorderly and irregular instances.

It evaluates the downward branches in accordance with various node properties, compares attributes across internal nodes of the decision tree in recursively, top-down manner and derives a conclusion from the decision tree's leaf nodes. The entire tree, then, represents a set of disjunctive expression rules, and each node along the way represents a conjunctive rule from root to leaf [12].

Consider the decision tree to be a Boolean operation. The function's input is the situation's object or all of its properties, and its output is a decision value of "yes" or "no." Each tree node in the decision tree represents a property test, each leaf node a Boolean value, and each branch one of the potential values for the testing attribute.

H. Random Forest Algorithm:

Random Forest developed by Leo Breiman [16] is a group of un-pruned classification or regression trees. It is called random as it selects the samples from training data randomly. Bootstrapped similar-sized datasets to the training set, or CART, are used to train an ensemble of Classification and Regression Trees (CART) that make up the RF. A collection of bootstraps that exclude any specific Once the tree has been built, records from the initial dataset are utilized as the test set. Those methods such as Bagging or Random subspaces [13,14], though it has composition of different classifiers and also has random property, have proven efficiency in performance.

For verification of the results Weka tool is used. Weka is a collection of machine learning algorithms. It has tools for mining association rules, clustering, regression, classification, and visualisation of data. It includes a number of algorithms and visualisation tools for data analysis and predictive

modelling, along with a graphical user interface. Weka supports a number of common data mining operations, including feature selection, data pre-processing, clustering, classification, and regressing. In the WEKA data mining tool, decision trees are generated using the J48, or Open Source Java, implementation of C4.5 release [17]. This Decision Tree algorithm is typical. Decision Tree Induction is one of the classification algorithms used in data mining. From the pre-classified data set, a model is inductively learned using the Classification algorithm [18].

I. Dataset

The dataset referred is simple, having all the statistical parameters. In manufacturing industry the components get manufactured are termed as "part" The parameters used are like part name, part number, part running time, cycle time with loading/unloading, actual production time, production per hour, expected quantity, achieved quantity etc. Different parts undergo different operation stages, so the parameter operation is further has sub-parameters.

Why did we use a decision tree rather than a random forest?

- When compared to random forests, decision trees are simpler. In contrast to a random forest, which mixes numerous decision trees, a decision tree combines certain decisions.
- As a result, the procedure is gradual yet lengthy. A decision tree, especially a linear one, runs quickly and without difficulty on big data sets.
- Choice trees are graphs that show all potential outcomes of a decision using a branching technique, which is a key distinction between them and the random forest algorithm.
- In contrast, a series of decision trees that follow the output are produced by the random forest method.

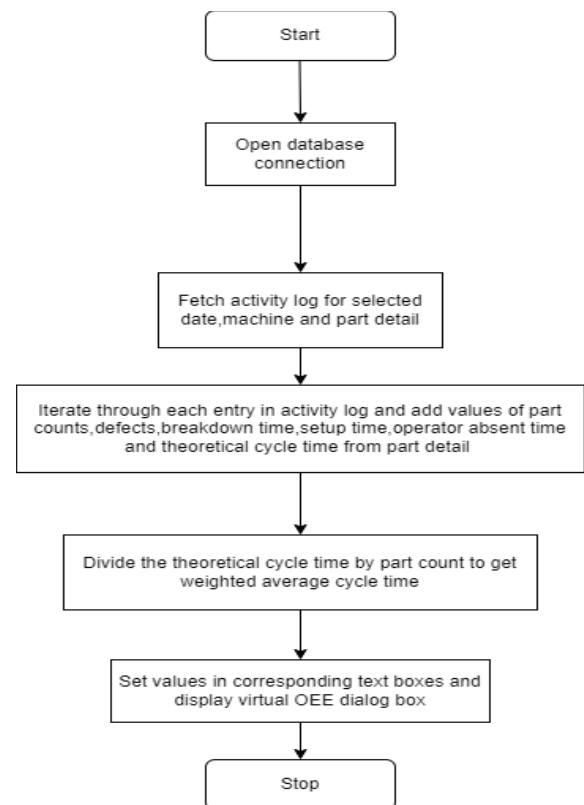


Fig. 3. Flowchart for OEE calculation

OEE dashboard works as shown in Fig 3, the dynamically updated dataset is referred by establishing connection to database. Selection of the particular machine parts and slot is done for which performance need to be calculated. According to weightage further analysis is displayed over dashboard

V. RESULTS AND DISCUSSION

The results we able receive through tool are shown in Fig. 5 and Fig. 6. Fig 5 shows how dashboard visualize the machine performance based on parts manufactured with all factors taken into consideration and Fig 6 visualize overall OEE performance.



Fig. 4. Performance tab in Dashboard



Fig. 5. OEE calculation in Dashboard

There are few limitations as listed below

- OEE and profitability are not correlated in any way.
- OEE shouldn't be regarded as a catch-all.
- OEE shouldn't be used to monitor performance outside of manufacturing because it isn't multi-dimensional enough for that.
- OEE variables need to be examined individually to identify actual areas for improvement.
- OEE variables need to be inspected individually to identify real areas for improvement

Some of the limitations overcome by another recent analytical dashboard [19, 20]. AI-ML can overcome the limitations and make the dashboard more informative.

VI. CONCLUSION

A BI solution is being created to support a single typical CNC machining Centre. It was intended to consolidate the data from CNC controllers in order to process all data and generate information on the machine's activity status. The

implementation of such a system will give many partners access to relevant information that will support decision-making.

The efficiency with which information is communicated to decision-makers can also be improved by centralising the various information sources and the formats they use (Excel files, relational databases, etc.). This can improve data modelling, data processing, information quality and quantity, as well as information quality and quantity. As a result, the requested and essential information is constantly available to all authorized users of the organization.

Due to the distinct requirements of various operational areas, the data files entered into Power BI as a source for computing specific indicators are subject to structural differences.

It must be suggested that Power BI be directly linked to this database in order to enable pre-planned and automatic data refreshes. As a result, the system update process will take less time.

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